

3.1 The Ground Up

Before we begin, there are a few things that need dealing with. Most importantly, you should read the previous chapter on HTML—you'll need all that knowledge to know what's happening inside your JavaScript code blocks. Next, we deal with some basic programming concepts.

How JavaScript works

JavaScript is an *Interpreted Language*—your browser reads each instruction and processes it in realtime, as opposed to compiled languages such as C and C++, where all the code you write needs to be translated into a language that your hardware can understand (called machine language) before it can be executed.

Variables

Variables store any information that you want to use in your programs. As the name suggests, you can change their value at any time. All variables need a *declaration*—you have to tell the interpreter what to treat as a variable. Let's take a simple example—a program that adds two numbers. Note that this is not actual code, but a representation of what it'll be doing.

JavaScript and Java

A near-foolproof way to get programmers to blow their gaskets is to draw no distinction between JavaScript and Java. Despite sharing four letters of the name, the two languages have very little in common! Java is a much more powerful and complex language than JavaScript, for one. It belongs to the same category of languages as C++ and C, languages we are most likely to see desktop and server applications written in. JavaScript, however, is a language meant only to introduce an element of programming to Web pages, nothing more.

The Declarations

A is a variable

B is a variable